

Residual-Basic Charges and Simultaneous Horizon First Laws in Static Weyl Gravity

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Abstract

Static spherical solutions of four-dimensional Weyl gravity possess a residual local $\text{Diff} \times \text{Weyl}$ freedom that changes the normalization of the chart Killing vector. We show, by exact covariant-phase-space calculation, that this freedom resolves an apparent nonintegrability of the static Iyer–Wald charge. On the Mannheim–Kazanas component of a Laurent-complete Bach-flat family, the bare charge associated with ∂_t is not closed. Requiring the field-dependent generator and its charge to descend to the residual quotient forces

$$\chi = uf(J)\partial_t, \quad u = \beta(2 - 3\beta\gamma),$$

where J is the single continuous residual invariant. For the canonical choice $f = 1$, the corrected charge is exact, with Hamiltonian

$$H = -16\pi\alpha\beta^2(9\beta^2\gamma^2k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k).$$

At every simple root r_h of the metric function, the Wald entropy and signed temperature

$$S_h = 64\pi^2\alpha\beta\frac{2 - 3\beta\gamma + \gamma r_h}{r_h}, \quad T_h = \frac{uB'(r_h)}{4\pi}$$

obey $dH = T_h dS_h$. The same one-form dH therefore gives simultaneous first laws at all simple horizons, including an exact non-Einstein three-horizon example. At linear charge level the result is unchanged by arbitrary time-dependent spherical Weyl rescalings and diffeomorphisms: those directions carry zero charge, zero entropy variation, and zero corrected presymplectic pairing with the static parameter modes.

The theorem is deliberately scoped. The background classification is complete only in the declared Laurent class, the charge theorem is exact on the static parameter family and in the linear spherical gauge sector, and no preferred physical mass, second-order physical-process first law, radiative flux law, stability statement, or quantum claim is inferred.

AI authorship and computational accountability

GPT-5.6.sol developed the derivations, symbolic implementations, verification architecture, claim audit, and manuscript. Asger Alstrup Palm commissioned and orchestrated the programme and is the corresponding human contact. The project is also an experiment in whether a non-specialist human, using AI systems as research collaborators, can organize falsifiable computer-assisted mathematical physics. Every promoted identity in this paper is bound to a machine-readable exact certificate and to a separately implemented verifier. The human orchestrator has not independently checked every tensor calculation; the manuscript therefore remains subject to expert review.

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1 Question, contribution, and claim boundary

For a stationary black hole in an asymptotically flat theory, the normalization of the horizon generator is usually fixed by the asymptotic clock. Generic static spherical solutions of pure Weyl gravity are not asymptotically flat, and their Schwarzschild-gauge representatives retain a local radial-diffeomorphism/Weyl freedom. The chart vector ∂_t is consequently not a residual-basic object.

This matters already on the three-dimensional static parameter family. The Iyer–Wald surface form for ∂_t is radially conserved but not integrable in parameter space. Adding a parameter-local corner function cannot cure this, because it cannot change the exterior derivative of the charge one-form. Our central observation is that the obstruction disappears when the generator is normalized as an object on the residual quotient. Basicness determines the normalization up to a function of the sole quotient invariant, and the resulting charge is closed.

The static Bach-flat family itself is classical [2, 3]; conformal-gravity black-hole thermodynamics and additional spin-two hair have also been studied in asymptotically AdS and Einstein–Weyl settings [6]. The contribution here is not a new solution family. It is:

1. an exact residual-quotient derivation of the field-dependent normalization;
2. an exact Hamiltonian and Wald entropy in that normalization;
3. one first-law identity holding simultaneously at every simple horizon;
4. a linear spherical audit showing that the charge is not selected by a conformal frame or by a spherical coordinate gauge.

Lü, Pang, Pope, and Vázquez-Poritz use an AdS ensemble with an independent thermodynamic pair for massive spin-two hair [6]. Our one-term identity is instead a quotient statement on a restricted

static family with no preferred asymptotic clock. The two constructions therefore answer different normalization and ensemble questions.

1.1 Dependency closure

Table 1 separates the exact theorem chain from open extensions. The dependency tags are those used in the computational archive. In the compact status column, LA means LOCAL-ALGEBRAIC and RM means REDUCED-MODE; the abbreviated certificate names are expanded in Section 9.

Result	Evidence	Status	Role in main theorem
Laurent Bach-flat locus, residual action, horizon fixture	BH0	LA, exact	Background and quotient geometry
Bare static Iyer–Wald form	BH1	LA+RM, exact	Exhibits the normalization obstruction
Basic generator, H, S, T , simultaneous first law	BH1A	LA+RM, exact	Main static theorem
Spherical conformal/diffeomorphism audit	BH1B	LA+RM, exact at linear charge level	Frame and gauge independence
Radiative flux, physical-process law, stability	none	open	Not used and not claimed

Table 1: Logical dependency of the paper. “Exact” means symbolic rational or polynomial identity, not floating-point evidence.

2 Conventions and the static Laurent family

We use signature $(-+++)$ and action

$$S_W[g] = \alpha \int_M \sqrt{-g} C_{abcd} C^{abcd} d^4x. \quad (1)$$

Our Bach-tensor convention is

$$B_{ab} = \nabla^c \nabla^d C_{acbd} + \frac{1}{2} R^{cd} C_{acbd}. \quad (2)$$

The field equation is $B_{ab} = 0$.

Start from the areal-radius ansatz

$$ds^2 = -a(r)dt^2 + b(r)dr^2 + r^2 d\Omega_2^2. \quad (3)$$

Before conformal gauge fixing, the exact Bach tensor is diagonal, trace-free, divergence-free, and has two independent components. We then choose $b = 1/a$, write $a = B$, and restrict to

$$B(r) = w - \frac{u}{r} + \gamma r - kr^2 + \frac{c_2}{r^2} + c_3 r^3. \quad (4)$$

Theorem 2.1 (Bach-flat locus in the declared Laurent class). *The metric (4) is Bach-flat if and only if*

$$c_2 = c_3 = 0, \quad w^2 + 3u\gamma = 1. \quad (5)$$

Proof. Substitution of (4) into the two independent Bach rows, clearing the declared denominators and equating Laurent coefficients, gives an exact polynomial ideal. Its reduced Gröbner basis is

$$\{c_2, c_3, w^2 + 3u\gamma - 1\}.$$

This proves necessity and sufficiency within the ansatz. The producer and independent curvature implementation both recompute the vanishing tensor; mutating the correlated constant term or adding an r^{-2} tail produces explicit nonzero Bach components. $\square$

Remark 2.2 (Scope of completeness). Theorem 2.1 is not presented as a new proof of the full Birkhoff–Riegert theorem [2]. Generalized Birkhoff statements and nonstandard spherical branches require additional global and nondegeneracy hypotheses [7]. Here “complete” always means complete in (4).

The Mannheim–Kazanas chart is

$$B(r) = 1 - 3\beta\gamma - \frac{\beta(2 - 3\beta\gamma)}{r} + \gamma r - kr^2, \quad (6)$$

so

$$u = \beta(2 - 3\beta\gamma), \quad w = 1 - 3\beta\gamma. \quad (7)$$

It parametrizes the component through $w = 1$ used in the charge theorem.

Proposition 2.3 (Correct Einstein locus). *On the full Laurent locus (5), the metric is Einstein in the areal-radius Schwarzschild chart if and only if*

$$\gamma = 0, \quad w = 1. \quad (8)$$

On the Mannheim–Kazanas component (6), this reduces to $\gamma = 0$.

Proof. On the full locus the scalar curvature is

$$R = 12k - \frac{6\gamma}{r} + \frac{2(1 - w)}{r^2}. \quad (9)$$

An Einstein metric has constant scalar curvature, forcing $\gamma = 0$ and $w = 1$. Direct substitution then gives $R_{ab} = 3kg_{ab}$. In the MK chart, the trace-free Ricci defect contains

$$E_{\theta\theta} = -\frac{\gamma}{2}(r - 3\beta),$$

so $\gamma = 0$ is necessary and sufficient there. The allowed Laurent point $(\gamma, w) = (0, -1)$ is therefore Bach-flat but not Einstein; it belongs to a complementary Kantowski–Sachs-type chart rather than to the MK component through $w = 1$. $\square$

3 Residual quotient and its invariant

Put $x = 1/r$ and

$$Q(x) = -ux^3 + wx^2 + \gamma x - k. \quad (10)$$

The roots of Q encode the nonzero roots of B . The form-preserving local transformation

$$r = \frac{\rho}{1 - c\rho}, \quad g \mapsto (1 - c\rho)^2 g \quad (11)$$

acts as the translation $Q(x) \mapsto Q(x - c)$. Equivalently,

$$\begin{aligned} u &\mapsto u, & w &\mapsto w + 3cu, \\ \gamma &\mapsto \gamma - 2cw - 3c^2u, & k &\mapsto k + c\gamma - c^2w - c^3u. \end{aligned} \quad (12)$$

A dilation acts by

$$(\beta, \gamma, k) \mapsto (\beta/\lambda, \lambda\gamma, \lambda^2k), \quad Q(x) \mapsto \lambda^2Q(x/\lambda). \quad (13)$$

The conformal factor in the c -map may vanish at finite radius, so (12) is used as a local quotient statement, not as a global identification of black-hole exteriors.

On the MK parameters the infinitesimal generators are

$$X_c = -3\beta^2\partial_\beta + (6\beta\gamma - 2)\partial_\gamma + \gamma\partial_k, \quad (14)$$

$$X_\lambda = -\beta\partial_\beta + \gamma\partial_\gamma + 2k\partial_k. \quad (15)$$

They have generic rank two. A continuous invariant is

$$J = u^2 \text{disc}(Q). \quad (16)$$

Translation invariance of the discriminant and its scaling weight prove $X_c J = X_\lambda J = 0$. Moreover, $\text{disc}(Q) = 0$ is precisely the multiple-root, hence degenerate-horizon, locus. Locally on a regular orbit component, every continuous invariant is therefore a function of J .

4 The bare charge is not integrable

For a diffeomorphism-covariant Lagrangian depending algebraically on the Riemann tensor, define

$$E^{abcd} = \frac{\partial(\alpha C_{efgh} C^{efgh})}{\partial R_{abcd}} = 2\alpha C^{abcd}.$$

We use the Iyer–Wald representative [4, 5]

$$\theta^a(g; \delta g) = 2 \left(E^{abcd} \nabla_d \delta g_{bc} - \delta g_{bc} \nabla_d E^{abcd} \right), \quad (17)$$

$$Q_\xi^{ab} = -E^{abcd} \nabla_c \xi_d + 2\xi_d \nabla_c E^{abcd}. \quad (18)$$

The sphere conventions are

$$(Q_\xi)_{\theta\phi} = 2\sqrt{-g} Q_\xi^{tr}, \quad (i_\xi \theta)_{\theta\phi} = -\sqrt{-g} \xi^t \theta^r.$$

With $\xi = \partial_t$, integration over any sphere gives the parameter-space one-form

$$\mathcal{F} = F_\beta d\beta + F_\gamma d\gamma + F_k dk, \quad (19)$$

where

$$\begin{aligned} F_\beta &= 16\pi\alpha(12\beta\gamma k - \gamma^2 - 4k), \\ F_\gamma &= 16\pi\alpha\beta(6\beta k - \gamma), \\ F_k &= -16\pi\alpha\beta(2 - 3\beta\gamma). \end{aligned} \quad (20)$$

Each component is exactly independent of the sphere radius. As a normalization control, replacing E^{abcd} by the Einstein–Hilbert tensor reproduces $16\pi \delta m$ on Schwarzschild and Schwarzschild–de Sitter in the convention $16\pi G = 1$.

Proposition 4.1 (Bare nonintegrability). *The chart-normalized form $\mathcal{F}$ is not closed:*

$$d\mathcal{F} = 16\pi\alpha [\gamma d\beta \wedge d\gamma + 2(1 - 3\beta\gamma) d\beta \wedge dk - 3\beta^2 d\gamma \wedge dk]. \quad (21)$$

It obeys

$$i_{X_c}\mathcal{F} = 0, \quad i_{X_c}d\mathcal{F} = 0, \quad (22)$$

$$i_{X_\lambda}\mathcal{F} = 0, \quad i_{X_\lambda}d\mathcal{F} = \mathcal{F}. \quad (23)$$

Consequently no parameter-local boundary function can make $\mathcal{F}$ integrable.

Proof. Equations (21)–(23) follow by exact differentiation of (20). If a boundary function W were added, the one-form would change by dW , leaving $d\mathcal{F}$ unchanged. The kernel of $d\mathcal{F}$ is the X_c direction at generic points. Thus quotienting only the translation leaves a nondegenerate two-form on the remaining two-dimensional parameter quotient; the obstruction is not removed merely by discarding that proper-gauge direction. $\square$

5 Basicness fixes the field-dependent generator

For a field-dependent generator $\chi = N(\beta, \gamma, k)\partial_t$, we use the corrected surface form

$$k_\chi(g; \delta g) = \delta Q_\chi - Q_{\delta\chi} - i_\chi\theta(g; \delta g). \quad (24)$$

Because N is spacetime-constant on each static solution and the Noether charge is linear in χ , the $Q_{\delta\chi}$ term cancels all dN contributions. The corrected charge is therefore

$$\delta H_\chi = N\mathcal{F}. \quad (25)$$

Definition 5.1. A parameter-space form is *residual-basic* if it is horizontal and invariant under both vector fields (14) and (15).

Theorem 5.2 (Residual-basic normalization). *On a connected regular component with $u \neq 0$, the corrected charge $N\mathcal{F}$ is residual-basic only if*

$$N = uf(J) \quad (26)$$

for a function f of the quotient invariant. Conversely, every normalization (26) is basic and integrable. For $f = 1$,

$$u\mathcal{F} = dH, \quad H = -16\pi\alpha\beta^2 D_2, \quad (27)$$

where

$$D_2 = 9\beta^2\gamma^2 k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k. \quad (28)$$

Proof. Cartan's identity and (22) give $\mathcal{L}_{X_c}\mathcal{F} = 0$. Hence invariance of $N\mathcal{F}$ requires $X_c N = 0$. Similarly, (23) gives $\mathcal{L}_{X_\lambda}\mathcal{F} = \mathcal{F}$, so invariance requires $X_\lambda N = -N$. Directly,

$$X_c u = 0, \quad X_\lambda u = -u.$$

Since the residual orbit has codimension one and J is its local invariant, the general local solution is $N = uf(J)$.

Exact differentiation of (27) gives $dH = u\mathcal{F}$, proving closure for $f = 1$. The same form is horizontal under both residual fields. For general $f(J)$, $f(J)dH$ is closed because the exact identity $dH \wedge dJ = 0$ holds. It is therefore locally the differential of a reparametrized quotient Hamiltonian. $\square$

The discriminant factorization sharpens the quotient interpretation. Put

$$D_1 = 27\beta^2 k - 3\beta\gamma - 1. \quad (29)$$

Then

$$J = -u^2 D_1 D_2. \quad (30)$$

The Hamiltonian vanishes on the $D_2 = 0$ component of the degenerate-horizon locus. On the Einstein control family $\gamma = 0$,

$$H = -64\pi\alpha\beta^2 k. \quad (31)$$

On the Schwarzschild subfamily $k = \gamma = 0$, it vanishes identically.

Remark 5.3 (Normalization versus physical mass). Equation (27) defines the Hamiltonian of the residual-basic generator on the declared static phase-space slice. We do not call it a universal physical mass. Multiplication by $f(J)$, component orientation, and any future choice of physical asymptotic clock change the normalization. On a fixed-falloff ensemble with γ and k fixed, the nontrivial residual transformations are generically frozen, but this does not create a preferred clock on the full family.

6 Entropy and simultaneous first laws

Let r_h be any simple positive root of B . The Wald entropy is

$$S_h = -2\pi \int_{S_{r_h}} E^{abcd} \epsilon_{ab} \epsilon_{cd} dA. \quad (32)$$

With $E^{abcd} = 2\alpha C^{abcd}$ and $\epsilon_{tr} = 1$, exact evaluation gives

$$S_h = 64\pi^2 \alpha \beta \frac{2 - 3\beta\gamma + \gamma r_h}{r_h}. \quad (33)$$

For the algebraic orientation $\chi = u\partial_t$, define the signed surface-gravity temperature

$$T_h = \frac{\kappa_\chi}{2\pi} = \frac{uB'(r_h)}{4\pi}. \quad (34)$$

This is the signed quantity entering the oriented first law; a positive Hawking temperature would use $|uB'(r_h)|/(4\pi)$.

Theorem 6.1 (Simultaneous horizon first laws). *For every simple root r_h of (6), the functions (27), (33), and (34) satisfy*

$$\boxed{dH = T_h dS_h} \quad (35)$$

in all three parameter directions. If one solution has simple roots $r_1, \dots, r_n$, then

$$T_1 dS_1 = \dots = T_n dS_n = dH. \quad (36)$$

Proof. Along the horizon locus $B(r_h) = 0$, any parameter $p \in \{\beta, \gamma, k\}$ obeys

$$\frac{dr_h}{dp} = -\frac{\partial_p B(r_h)}{B'(r_h)}. \quad (37)$$

Use this in the total derivative of (33), subtract it from $\partial_p H/T_h$, clear denominators, and reduce the numerator modulo

$$r_h B(r_h) = 0. \quad (38)$$

The remainder is identically zero for each of the three parameters. This calculation uses no choice of root beyond simplicity, proving the result at all roots simultaneously. $\square$

For Schwarzschild, $r_h = 2\beta$, so

$$S_h = 64\pi^2\alpha$$

is independent of β , consistently with $H = 0$ on that subensemble. No pressure–volume or extra source term is needed for (35) on the declared residual-basic static family. This does not exclude such terms in a different ensemble with independently varied boundary data.

6.1 Exact non-Einstein three-horizon fixture

Consider

$$(\beta, \gamma, k) = \left(\frac{3}{2}, \frac{12}{19}, \frac{1}{19}\right). \quad (39)$$

Then

$$rB(r) = -\frac{1}{19}(r-1)(r-3)(r-8), \quad u = -\frac{24}{19}, \quad (40)$$

and the trace-free Ricci witness is

$$E_{\theta\theta} = -\frac{3(2r-9)}{19} \neq 0.$$

The metric is smooth at all three roots in ingoing Eddington–Finkelstein coordinates, and all certified scalar curvature invariants are finite there. In the algebraic $N = u$ orientation,

$$H = -\frac{14400\pi\alpha}{6859} \quad (41)$$

and

r_h	S_h	T_h
1	$-384\pi^2\alpha/19$	$84/(361\pi)$
3	$640\pi^2\alpha/19$	$-20/(361\pi)$
8	$960\pi^2\alpha/19$	$105/(1444\pi)$

The three identities $T_h dS_h = dH$ hold exactly in every parameter direction. Since $u < 0$ on this component, choosing the opposite future-directed orientation flips H and every signed T_h , leaving the first laws unchanged.

7 Linear spherical frame and gauge audit

The preceding derivation lives on the static parameter slice. It is nevertheless important to decide whether its normalization secretly selects a conformal frame. We therefore consider the complete linear spherical gauge sector around both symbolic Schwarzschild and the non-Einstein fixture:

$$\delta_\omega g_{ab} = 2\omega(t, r)g_{ab}, \quad (42)$$

$$\delta_\xi g_{ab} = \mathcal{L}_\xi g_{ab}, \quad \xi = a(t, r)\partial_t + b(t, r)\partial_r. \quad (43)$$

Theorem 7.1 (Linear spherical gauge independence). *At linear charge level:*

1. $k_\chi(2\omega g) = 0$ componentwise for arbitrary $\omega(t, r)$;
2. the entropy variation under $2\omega g$ vanishes on the symbolic MK family;

3. the corrected presymplectic current between a conformal variation and any static parameter mode vanishes;
4. $k_\chi(\mathcal{L}_\xi g) = 0$ for arbitrary spherical ξ ;
5. the extension of $N = u$ is uniquely fixed by $\delta N = du$ on parameter directions and $\delta N = 0$ on spherical conformal and diffeomorphism directions.

Thus no boundary clock or preferred conformal representative is needed for the static theorem at this linear charge level.

Proof. The conformal statements follow from direct substitution in (24), exact conformal covariance of the Bach equation, and the literal variation of the density $\sqrt{-g}\theta^a$. The density variation is essential: omitting the trace contribution spoils current conservation. The diffeomorphism statement is verified through the on-shell Noether identity

$$\theta(\mathcal{L}_\xi g) - i_\xi(L\epsilon) = dQ_\xi \quad (44)$$

and the background identity obtained from its commutator with χ . Both routes are independently evaluated in the certificate.

For uniqueness, the bare Noether aperture is nonzero on the fixture. Any nonzero δN on a certified gauge direction would therefore assign that direction a nonzero charge, contradicting the componentwise annihilation above. The parameter variation of N is already fixed by Theorem 5.2. $\square$

The theorem is complete for the $\ell = 0$ dynamical gauge sector at linear charge level. It does not compute the bilinear radiative flux for $\ell \geq 2$, and it does not imply a second-order physical-process first law.

8 Ensembles, interpretation, and non-results

The c -transformation preserves the fixed-falloff data $\{\gamma, k\}$ only for $c = 0$, away from the algebraic locus

$$u(w^2 - 3) = 0.$$

The $u = 0$ component is excluded because the basic normalization degenerates. The dilation preserves the same ensemble only at $\lambda = 1$, except on the Schwarzschild subensemble $\gamma = k = 0$, where it acts freely and consistently with $H = 0$ and constant entropy. On a finite exterior $[r_h, r_{\text{out}}]$, the c -map conformal factor is smooth and positive precisely when $1 + cr > 0$ throughout that interval.

These observations support a precise interpretation:

The nonintegrability of the chart-normalized charge is a normalization artifact. Requiring the generator and charge to descend to the complete local residual quotient selects an integrable class of generators. It does not, by itself, choose a unique physical clock.

The following statements are not established:

- completeness of static Bach vacua beyond the Laurent ansatz;
- a preferred physical matter or clock conformal frame;
- a universal physical mass valid across inequivalent boundary ensembles;

- a nonlinear or second-order physical-process first law;
- the bilinear radiative symplectic flux through a horizon or infinity;
- stability, quasinormal ringing, Hawking radiation, or any quantum property.

9 Computational evidence and reproduction

The calculations are computer-assisted but exact. Tensor components, polynomial reductions, and charge identities are represented by symbolic rational functions. No theorem in this paper is inferred from a decimal approximation. Producer and verifier use separately written curvature pipelines; the verifier constructs the Weyl tensor through an independent Schouten/Kulkarni–Nomizu route. Mutation tests ensure that certificate-like but false inputs are rejected.

Certificate	Reproduction and independent verification
BH0_STATIC_SPHERICAL_BACKGROUND.json	python3 black_hole_programme/bh0_background.py python3 black_hole_programme/verify_bh0_background.py
BH1_LEE_WALD_PREFLIGHT.json	python3 black_hole_programme/bh1_lee_wald_preflight.py python3 black_hole_programme/verify_bh1_lee_wald_preflight.py
BH1A_NORMALIZED_GENERATOR.json	python3 black_hole_programme/bh1a_normalized_generator.py python3 black_hole_programme/verify_bh1a_normalized_generator.py
BH1B_DYNAMICAL_EXTENSION.json	python3 black_hole_programme/bh1b_dynamical.py python3 black_hole_programme/verify_bh1b_dynamical.py

Table 2: Authoritative certificate families. Re-running a producer is reproduction; the separately implemented verifier is the independent check.

The BH0 mutations (i) remove the correlated $-3\beta\gamma$ constant and (ii) add an r^{-2} tail to the three-horizon fixture; both yield nonzero Bach components. The bare-charge computation includes Einstein normalization controls and retains $d\mathcal{F} \neq 0$ as a negative control. The BH1A verifier independently recomputes entropy and reduces the first-law polynomial at all three fixture roots. The BH1B verifier recomputes the conformal and diffeomorphism annihilation identities.

10 Conclusion

Static Weyl gravity does not supply a preferred Killing normalization merely by presenting a Schwarzschild-gauge metric. On the certified Mannheim–Kazanas family, treating ∂_t as fixed produces a radially conserved but nonintegrable Iyer–Wald form. Treating the generator as field dependent and demanding residual basicness instead forces $\chi = uf(J)\partial_t$. Its charge is integrable, its Wald entropy is explicit, and one Hamiltonian variation satisfies a signed first law at every simple horizon simultaneously.

The result is stronger than a coordinate identity and weaker than a complete black-hole thermodynamics. It is invariant under the complete linear spherical conformal and diffeomorphism gauge sector, but it does not fix a physical asymptotic clock or control nonlinear radiative flux. Those boundaries are part of the theorem, not deferred qualifications.

References

- [1] R. Bach, *Zur Weylschen Relativitätstheorie und der Weylschen Erweiterung des Krümmungstensorbegriffs*, Math. Z. **9** (1921) 110–135.
- [2] R. J. Riegert, *Birkhoff’s theorem in conformal gravity*, Phys. Rev. Lett. **53** (1984) 315–318.
- [3] P. D. Mannheim and D. Kazanas, *Exact vacuum solution to conformal Weyl gravity and galactic rotation curves*, Astrophys. J. **342** (1989) 635–638.
- [4] J. Lee and R. M. Wald, *Local symmetries and constraints*, J. Math. Phys. **31** (1990) 725–743.
- [5] V. Iyer and R. M. Wald, *Some properties of Noether charge and a proposal for dynamical black hole entropy*, Phys. Rev. D **50** (1994) 846–864, [arXiv:gr-qc/9403028](#).
- [6] H. Lü, Y. Pang, C. N. Pope, and J. F. Vázquez-Poritz, *AdS and Lifshitz black holes in conformal and Einstein–Weyl gravities*, Phys. Rev. D **86** (2012) 044011, [arXiv:1204.1062](#).
- [7] P. Jizba and T. Lehečková, *Generalized Birkhoff theorems and $2 + 2$ direct product spacetimes in Weyl conformal gravity*, Phys. Rev. D **113** (2026) 044060, [doi:10.1103/9g8r-lrjk](#), [arXiv:2512.18482](#).